



Title: Lightweight Interpretable GNN with API Abstraction for Android Malware Detection

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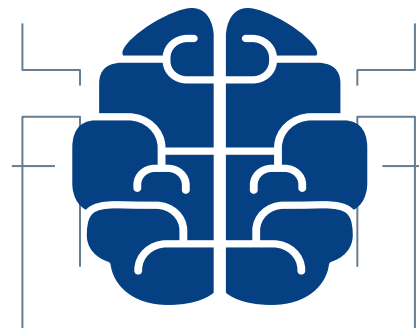
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Problem Statement

The Problem: Malware is Hiding in Plain Sight

- **Code Disguising:** Malware developers constantly change API names and "scramble" their code to bypass traditional security.
- **Literal Detection:** Existing systems look at exact names (like "sendSMS"). If the name is changed, the system fails.
- **The Gap:** There is a lack of focus on the meaning of the actions, making current antivirus tools fragile and easily fooled by new tricks.



Paper	Contributions	Year	Used Algorithm
GNN-based Android Malware Detection Framework	This framework addresses the problem of advanced hidden malware by extracting deep semantic meanings from code. It treats detection as a graph classification task to capture complex behavioral patterns.Methodology: Uses CodeBERT and TextCNN to create opcode embeddings, which are then processed using GraphSage and SAGPooling on an API call graph.	2022	GNN
Exploring GNN-LRP for Android Malware Detection	Focuses on the "Black Box" problem of AI by making GNN decisions interpretable. It identifies which specific code paths or sensitive APIs actually triggered the "malware" label.Methodology: Applies Layer-wise Relevance Propagation (GNN-LRP) to assign scores to nodes and edges within subgraphs generated by the MsDroid framework	2022	GNN
NF-GNN: Network Flow Graph Neural Networks	Moves away from looking at code to looking at network traffic. It models the entire communication flow between devices as a graph to detect malicious data-sharing patterns.Methodology: Extracts Network Flow Graphs from traffic data and uses a specialized GNN that treats edge features (connection metadata) as primary data for classification	2021	GNN
Ransomware Detection Based on Adaptive GNN Learning	Specifically designed to catch ransomware by identifying "global" relationships between distant API calls. It prevents the model from missing malicious steps that are hidden far apart in the code.Methodology: Utilizes a Text Graph Convolutional Network (TextGCN) with Adaptive Diffusion Convolution (ADC) to automatically learn the best connections between sensitive API nodes.	2023	GNN
Network Traffic Graph for 5G Mobile	Targeted at 5G and IoT environments where traffic is dense and complex. it identifies malware by analyzing the structural relationships between different traffic flows rather than just single data packets.Methodology: Converts pcap network packets into vector-based traffic graphs and processes them through a custom Graph Neural Network architecture designed for high-density IoT data.	2023	GNN



Proposed Methodology

Phase 1: Data Acquisition & Pre-processing

Dataset: Utilize the Drebin and AndroZoo datasets (containing both malicious and safe apps).

Static Analysis: Use tools like Androguard to unpack the APK and extract the Control Flow Graph (CFG) and API call sequences.





Proposed Methodology

Phase 2: The Core Innovation (API Abstraction)

Semantic Mapping: Instead of using raw API names (e.g. `getLine1Number()`), we map them to Behavioral Categories (e.g. `USER_IDENTITY`).

De-cluttering: This step removes "code-hiding" tricks and simplifies thousands of complex APIs into a few dozen meaningful actions.





Gaps in existing research

Existing models focus too literally on API names, making them fragile against modern hiding techniques (obfuscation). Furthermore, current graph-based systems are often too computationally heavy for mobile devices and act as 'black boxes' that fail to explain the behavioral intent behind their decisions.



Gap Analysis

Paper Title	Primary Approach	The GAP	Proposed Solution
GNN-based Android Malware Detection Framework	CodeBERT Semantics	Too heavy for mobile, relies on exact opcode sequences.	Lightweight API Categories (Fast & Efficient)
Exploring GNN-LRP for Android Malware Detection	Explainable AI (XAI)	Uses complex math, hard for normal users to understand.	Behavioral Labels (e.g. "Sends Data to Web") for clear logic.
NF-GNN: Network Flow Graph Neural Networks	Network Flow	Ignores the app's code, only sees data traffic.	Analyzes App Intent before the data is even sent.
Ransomware Detection Based on Adaptive GNN Learning	Global Correlation	Struggles with renamed/hidden API calls (obfuscation).	Abstraction Layer that sees the "meaning" not the name.
Network Traffic Graph for 5G Mobile	5G IoT complexity	Focuses on environment, not the specific malicious behavior.	Focuses on the Action Flow inside the application.



What's New? (Our Project Scope)

Our project introduces a Behavioral Abstraction Layer that translates scrambled API names into clear intent categories before they reach the AI. By using a lightweight GNN on these simplified patterns, we achieve a system that is not only harder to fool but also fast enough for mobile devices and easy for humans to understand.





Thank You

